Fact Omission vs. Rhetorical Framing: Disentangling the Predictors of Media Bias

Aditya Sai, Aayush Batra, Yash More

IIIT Hyderabad

April 2026

Motivation: How Do We Detect Media Bias?

Traditional bias detection focuses heavily on **lexical choices**:

- Word choice, sentiment, and emotional framing language.

However, bias also manifests through **structural choices**:

- **Selection Bias:** Which facts are included or omitted?
- **Framing Bias:** Where are those facts positioned within the discourse?

The Core Research Question

Is media bias better predicted by **fact omission** *(what is not said) or by the* **rhetorical positioning** *of shared facts (how it is structured)?*

We set out to rigorously test this by isolating both signals using Rhetorical Structure Theory (RST).

The Two-Tier Hypothesis of Media Bias

We hypothesize that media bias operates on two structural tiers:

Tier 1: Selection Bias (Omission)

The primary editorial decision. Sources choose *which* facts to report, creating coverage asymmetries.

- Example: A left-leaning source entirely omits a fact that a right-leaning source emphasizes.

Tier 2: Framing Bias (RST Structure)

The secondary editorial decision. For facts that *are* shared, the rhetorical positioning differs.

- We quantify this using **Rhetorical Structure Theory (RST)** depth and nuclearity.

Primer: Rhetorical Structure Theory (RST)

RST represents a document as a constituency tree over **Elementary Discourse Units (EDUs)** (typically clauses or sentences).

- **Nucleus:** The core, essential content of the text.
- **Satellite:** Supporting, background, or elaborative material.
- **Depth:** Nucleus EDUs closer to the root are structurally prominent. Satellite EDUs deep in the tree are structurally marginalized.

Dataset & Triplet Construction

- **Source Corpus:** 37,554 labeled news articles from AllSides.com.
- **Triplets:** We grouped articles covering the *exact same news event* into triplets: (1 Left, 1 Center, 1 Right) using SBERT cosine similarity.
- **Discourse Parsing:** We processed all articles using the `isanlp_rst_v3` neural parser to generate RST trees.

Final Usable Dataset:

- **1,748 Valid Triplets** (split securely into 80/10/10 Train/Val/Test with zero article leakage).
- **548,960 high-quality EDUs** retained after filtering.

Methodology: Fact Clustering

To identify shared facts across an event, we must group semantically identical EDUs.

Two-Stage Pipeline:

1. **SBERT Clustering:** Embed all EDUs in a triplet using `all-mpnet-base-v2` and apply Agglomerative Clustering (cosine distance, 0.35 threshold).
2. **Cross-Encoder Pruning:** To prevent "hallucinated" shared facts, we re-score all pairs using a `ms-marco` cross-encoder, dropping associations below 0.7.

Methodology: The Bipartite Approach

How do we compare structures across three different articles?

- We explicitly compare the Left and Right articles to the **Center article** (our baseline).
- For each fact cluster, we perform a **1-to-1-to-1 greedy matching** based on minimal depth differences.
- This yields two types of features...

Methodology: Extracting Features

1. Structural Features (δ_{str})

For EDUs that successfully match, we calculate the difference in rhetorical prominence (log-depth) between the biased source and the center.

2. Coverage Features (δ_{cov})

Unmatched EDUs generate explicit categorical signals representing omission:

- $\delta_{cov} = 0$: Fact is matched (shared).
- $\delta_{cov} = +1$: Biased source has the fact, Center omitted it.
- $\delta_{cov} = -1$: Center has the fact, Biased source omitted it.

Methodology: Vector Assembly

Each triplet yields two distinct Discourse Framing Index (DFI) vectors:

$$\mathbf{v}_{left} = [\delta^{(1)}_{cov,L}, \delta^{(1)}_{str,L}, \ldots, \delta^{(K)}_{cov,L}, \delta^{(K)}_{str,L}]$$
$$\mathbf{v}_{right} = [\delta^{(1)}_{cov,R}, \delta^{(1)}_{str,R}, \ldots, \delta^{(K)}_{cov,R}, \delta^{(K)}_{str,R}]$$

- Each vector serves as an independent sample for the classification task: *Predict if this vector represents a Left or Right divergence from Center.*

Results: Classification with All Fact Clusters

Evaluating the test split using the complete set of extracted clusters (which includes all coverage and omission signals):

Method	Test Accuracy
Bipartite Coverage + RF	**89.77%**
Bipartite Combined + RF	88.07%
Padded DFI + RF	87.78%
Padded DFI + SVM	67.05%

- **Random Forest (RF)** achieved nearly 90% accuracy, vastly outperforming SVMs (by +22.72%) by natively handling the discrete coverage deltas ($\{-1, 0, +1\}$).

The Feature Ordering Problem

Because each triplet describes a completely different news event, there is no universal semantic alignment for "Feature 1" across the dataset.

How do we order the fact clusters within the DFI vector?

- By minimum depth? (Shallowest facts first)
- By maximum prominence?
- By cluster size?

The Discovery: First-Mention Document Order

Surprisingly, explicitly sorting by RST depth or size **degraded performance**. The best strategy was simply preserving the chronological insertion order from our pipeline.

Why? The "Inverted Pyramid" of Journalism

Because our pipeline reads the articles linearly from top to bottom, the resulting dictionary natively sorts fact clusters by **their first chronological appearance in the text**.

In journalism, the most critical, headline-level facts are placed at the very beginning of the article. This organic narrative flow naturally captures the highest-signal omission events at the start of the feature vector!

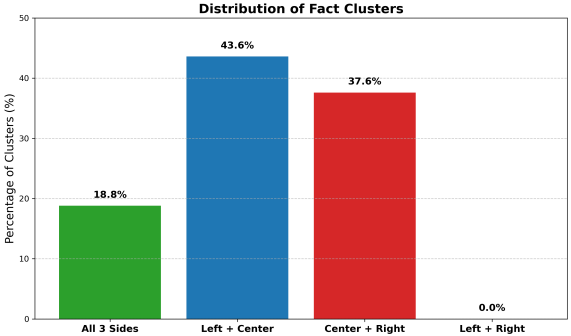
Experiment 3: Cluster Ordering Ablation Results

Ordering Strategy	Test Acc	Δ vs Baseline
First-Mention Document Order	**89.77%**	—
Min depth (shallowest first)	75.28%	-14.49%
Max prominence	74.15%	-15.62%
Cluster size	71.59%	-18.18%

Sorting clusters organically by their first document appearance outperforms explicit prominence-based reordering.

The "Siloed Realities" Discovery

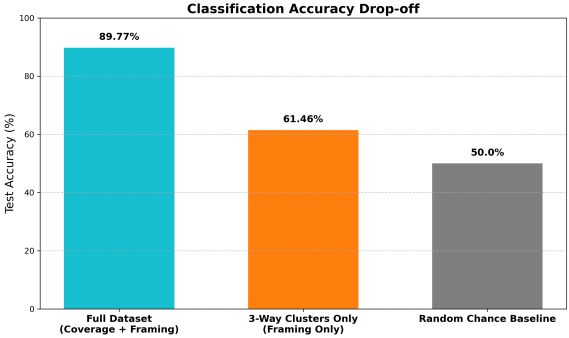
When we analyzed the clusters, we found a stark reality about modern media: **left and right sources rarely report the exact same facts.**



- Only **18.8%** of fact clusters exist in all three sources.
- Left and right share an average of only **3.95 facts** across an entire event.
- **24.4% of triplets (427)** have *zero* 3-way shared facts.

The 3-Way Experiment: Isolating Framing Bias

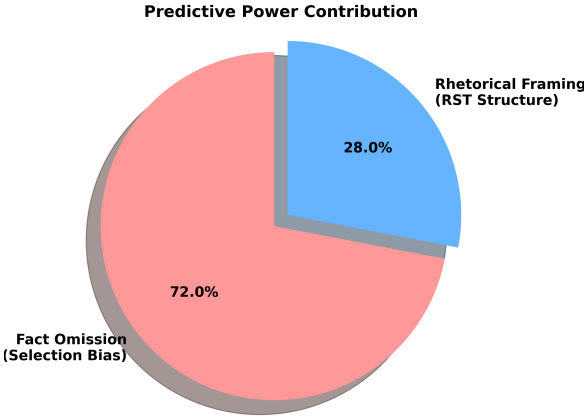
What happens if we *remove* the omission signal entirely? We created an isolated dataset containing **only facts reported by all three sides**.



- With omission removed, accuracy plummeted from **89.77%** to **61.46%**.
- This is barely above the **50% random chance baseline**.
- **Conclusion:** Framing alone is a surprisingly weak predictor of bias.

Quantifying Predictive Power: Omission vs. Framing

By comparing the performance drops from our 89.77% ceiling down to the 50% random baseline, we quantified the relative contribution of each tier:



- **Fact Omission (Selection Bias)** drives ∼**72%** of the predictive signal.
- **RST Structure (Framing Bias)** contributes only ∼**28%**.
- Our initial hypothesis—that RST structural positioning would be the main predictor—was proven incorrect.

The Need for Explainable AI in Journalism

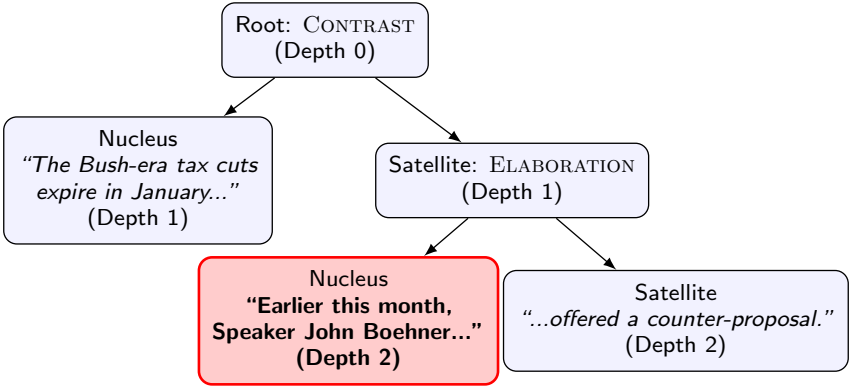
Deep learning bias classifiers (like Transformers) often function as black boxes. They predict bias based on latent lexical patterns, making it impossible to audit *why* an article was flagged.

Because our pipeline maps feature indices directly back to explicit strings (EDUs), we can leverage Random Forest **decision path tracking (Saabas method)** to generate human-readable explanations.

Explainability Demo: Tracing the Decision

Sample 1527: Predicting "Right-vs-Center"

The Right-leaning source **omitted** a fact that the Center source placed prominently in its discourse tree:



Because the Right source omitted the red highlighted EDU, our pipeline registered $f_0 = -1$, triggering a massive decision split in the Random Forest.

Explainability Demo: Feature Contributions

By tracking the Random Forest decision path, we can output exactly which omitted or added facts drove the final prediction logit:

Feature	Contribution	Mapped EDU Evidence (The "Why")
f0 = -1	+0.2486	**Center-only mention:** "Earlier this month, Speaker John Boehner ..." (Right omitted this)
f1 = -1	+0.0499	**Center-only mention:** "The Bush-era tax cuts expire ..." (Right omitted this)
f8 = +1	+0.0380	**Right-only mention:** "... automatic cuts to Pentagon spending." (Right added this)

Unlike a black box, we can definitively state: "The model classifies this as Right-leaning because it observes the omission of the Boehner/Tax cut facts and the inclusion of the Pentagon cut fact."

Discussion: The Two-Tier Model Validated

Our findings solidify a **two-tier model** for understanding and detecting media bias:

Tier 1 (Selection Bias): The Dominant Force

- Editorial selection of what facts to include or omit drives 72% of the signal.
- Media bias manifests primarily through distinct, siloed information bubbles rather than rhetorical spin on shared facts.

Tier 2 (Framing Bias): The Secondary Force

- Rhetorical positioning (depth and nuclearity) provides a weak but measurable signal for the rare facts that cross ideological boundaries.

Conclusion & Key Takeaways

1. **Omission dominates Framing:** Media bias is primarily about *what* is reported, not *how* it is structured.
2. **Siloed Realities:** Left and right sources cover the same events but rarely report the exact same facts (Avg 3.95 shared facts per event).
3. **Chronology is Salience:** The "First-Mention Document Order" naturally surfaces the most critical omission events due to journalistic inverted pyramid structures.
4. **Explainable AI in Journalism:** Our Bipartite Coverage approach (89.77% accuracy) is fully auditable, pinning down exactly which omitted or added sentence drove the prediction.

Thank You!
Questions?

Reproducibility

All code, data splits, and experimental configurations are open-source.

`https://github.com/itsadityasai/rhetorical-framing-auditor`

- **SBERT model:** `all-mpnet-base-v2`
- **Cross-encoder:** `ms-marco-MiniLM-L-6-v2`
- **Discourse Parser:** `isanlp_rst_v3`